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M. PHARM (PHARMACOLOGY)
(SEM II) THEORY EXAMINATION 2022-23
PHARMACOLOGICAL AND TOXICOLOGICAL SCREENING METHODS-II

Time: 3 Hours**Total Marks: 75****Note:** Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief. 10x2 = 20

- (a) What is toxicology and classify its type.
- (b) Define Ames test.
- (c) What is the difference between pharmacokinetics and toxickinetics?
- (d) Name any two OECD test guidelines for oral toxicity studies.
- (e) Write about the chromosomal abbreviation studies.
- (f) Define teratogenicity with teratogen as examples.
- (g) What is Comet assay?
- (h) Define HERG assay.
- (i) What is saturation kinetics?
- (j) Give the list of studies needed for IND submission.

SECTION B

2. Attempt any two parts of the following: 2x10 = 20

- (a) Give the general principles of good laboratory practice.
- (b) Write a short note on the following: -
 - (i) CNS safety pharmacology
 - (ii) *In-vivo* micronucleus assay
- (c) Explain in detail about the importance of regulatory toxicological studies.

SECTION C

3. Attempt any five parts of the following: 5x7 = 35

- (a) Discuss the role of ICH, OECD, and Schedule Y in toxicity studies.
- (b) What do you understand by Test item? Discuss the characterization for specific test items.
- (c) Write the objectives and principle of combined chronic toxicity/carcinogenicity studies.
- (d) Explain teratogenicity studies (segment II) in detail.
- (e) Write a descriptive note on concepts and importance of safety pharmacology.
- (f) Discuss the importance and applications of toxicokinetic studies.
- (g) Write a note on 'Alternative Methods to Animal Toxicity' testing.